14. Exploring Pro-Code Extension of a Zoho Creator Application Using Custom Functions and Zoho Catalyst

Aim

To explore how the Zoho Creator Job Application Management application can be extended using custom Deluge functions and Zoho Catalyst to support reusable business logic, external API integration, and advanced backend processing.

Theory

Zoho Creator provides low-code development features such as forms, reports, workflows, and Deluge scripting. When an application requires advanced processing or external integrations, custom functions and pro-code services can extend its capabilities.

1. Custom Deluge Functions

Concept: Custom Deluge functions are reusable scripts that perform specific tasks within a Zoho Creator application.

Application: In the recruitment application, the Generate_Offer_Letter_Function function checks a candidate ID and candidate information before preparing the offer-letter process.

Benefits:

- Reduces repeated code.
- Supports recruitment workflow automation.
- Improves code organization and reuse.
- Helps validate candidate information.

2. Zoho Catalyst Serverless Integration

Concept: Zoho Catalyst is a serverless development platform that supports building backend services using programming languages such as Java, Node.js, and Python.

Application: Catalyst could extend the recruitment application by processing candidate documents, generating offer-letter PDFs,

or handling advanced recruitment analytics through APIs.
Benefits:

- Supports custom backend development.
- Enables integration with external services.
- Allows complex processing to be handled outside Zoho Creator.
- Provides flexibility for future application enhancements.

Procedure

Step 1: Creating and Reviewing a Custom Deluge Function

1. Log in to Zoho Creator and open the Job Application Management application.
2. Navigate to the application settings or logic section.
3. Open Functions.
4. Locate the existing function Generate_Offer_Letter_Function.
5. Review its input argument, candidate_id, and the candidate validation logic.
6. Verify that the function checks the candidate ID and candidate information before proceeding.
7. Save any necessary changes and test the function using an appropriate candidate record.
8. Confirm that the function executes without errors.

Step 2: Exploring Zoho Catalyst

1. Open the Zoho Catalyst Console.
2. Sign in using an authorized Zoho account.
3. Explore the options for creating a Catalyst project.
4. Review the available serverless functions and supported programming languages.
5. Identify a possible recruitment-related use case, such as generating offer-letter PDFs or processing candidate documents.
6. If project creation is permitted, create a test project and examine its development options.

Step 3: Connecting Zoho Creator with Zoho Catalyst

1. Review the existing integration settings in Zoho Creator.
2. Check whether an appropriate connection to Zoho Catalyst has been configured.
3. If a connection is not available, explore the process of creating an authorized connection.
4. Identify a suitable Catalyst API endpoint for the selected recruitment task.
5. Review how a Deluge function can send data to an external endpoint using invokeurl.
6. Ensure that authentication details are stored securely.
7. Configure the integration only if the required Catalyst project, endpoint, and permissions are available.

Example Deluge structure for an API request:

```
response = invokeurl
[
    url : "YOUR_CATALYST_API_ENDPOINT"
    type : POST
    parameters : {"candidate_id": input.ID.toString()}
    connection : "catalyst_connection"
];
info response;
```

Note: Replace the endpoint and connection name with the actual configured values. This is an example structure, not a tested connection. The request parameters must match the API implemented in Catalyst.

Step 4: Testing and Validating the Workflow

1. Select a suitable test candidate record.
2. Execute the existing Generate_Offer_Letter_Function using the appropriate candidate ID.
3. Check the execution result and confirm whether the candidate validation succeeds.
4. If a Catalyst endpoint has been configured, send a test

request from Zoho Creator.

5. Verify that Catalyst receives the request and returns an appropriate response.
6. Check the execution logs for errors.
7. Confirm that the integration does not modify or delete existing candidate records unexpectedly.

Result

The use of custom Deluge functions for extending the Job Application Management application was explored. The existing Generate_Offer_Letter_Function was reviewed as an example of reusable recruitment logic, and Zoho Catalyst was examined as a potential platform for advanced backend processing and API integration. The Catalyst integration can be tested further when

a project, endpoint, and authorized connection are configured.